

Multi-Label Emotion Classification

Fine-Tuning BERT with PyTorch Lightning & Optuna

- Aakriti Dhakal

Project Scope

The Challenge

Goal: Detect complex, overlapping emotions in text.

Input: Raw Reddit comments.

Output: Probabilities for 5 distinct emotions.

Type: Multi-Label Classification (Not Binary).

The Dataset

Source: GoEmotions (58k comments)

Target Classes:

1. Joy
2. Sadness
3. Anger
4. Fear
5. Neutral

Project Scope

The Challenge

Goal: Detect complex, overlapping emotions in text.

Input: Raw Reddit comments.

Output: Probabilities for 5 distinct emotions.

Type: Multi-Label Classification (Not Binary).

The Dataset

Source: GoEmotions (58k comments)

Target Classes:

1. Joy
2. Sadness
3. Anger
4. Fear
5. Neutral

Baseline Architecture

Transfer Learning Approach

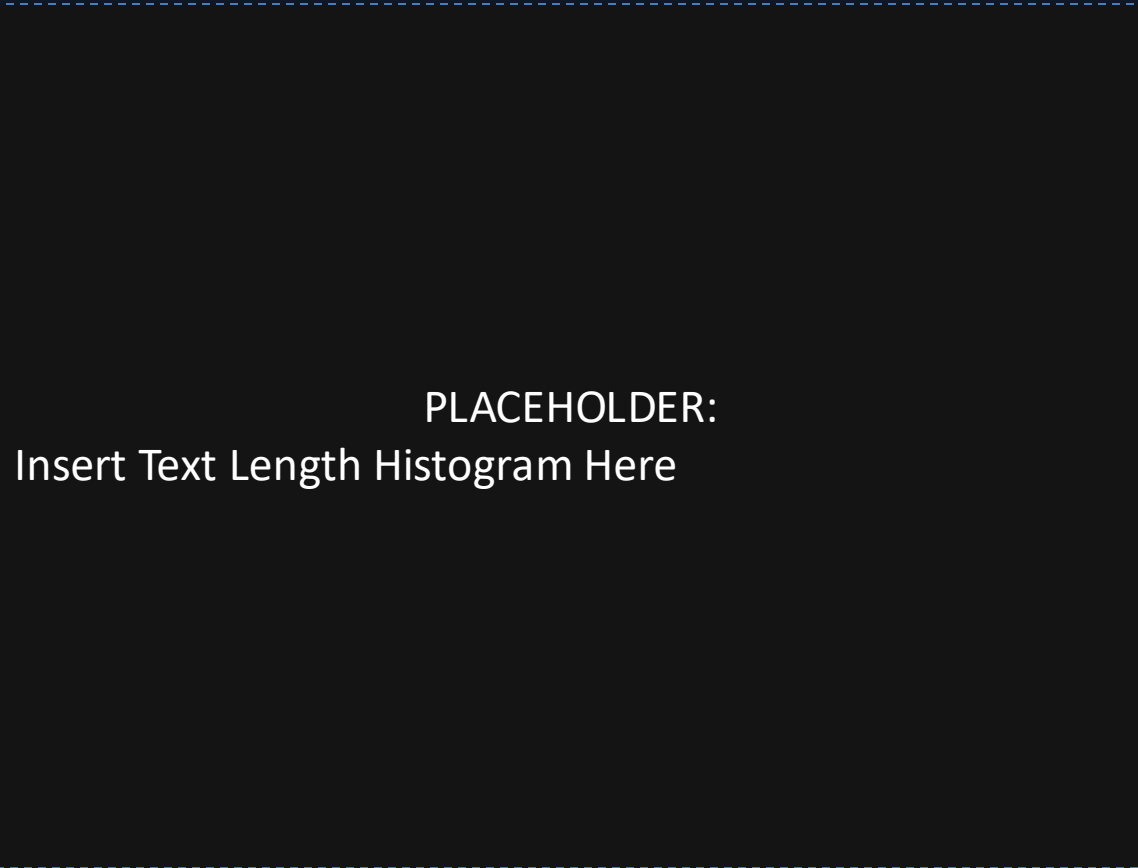
- Backbone: bert-base-uncased (12 Layers)
- Strategy: Unfrozen layers for full adaptation
- Pooling: [CLS] token representation
- Head: Linear Projection (768 -> 5)
- Activation: Sigmoid (Independent probabilities)
- Loss: BCEWithLogitsLoss

PLACEHOLDER:

Insert Architecture Diagram Here

(Input -> BERT -> Dropout -> Linear -> Sigmoid)

Data Pipeline & EDA



Preprocessing Strategy

Analysis Findings:

- Avg Length: 13 words
- 95th Percentile: 24 words
- Max Length: 33 words

Configuration:

- max_length = 48 (Covers 100% of data)
- Reduced memory usage by ~90% vs default 512 tokens.

Training Configuration

Framework

PyTorch Lightning

Handles loops, device placement, & checkpointing.

Optimizer

AdamW

Decoupled weight decay for better transformer generalization.

Scheduler

ReduceLROnPlateau

Monitors Val Loss. Halves LR if stagnant for 2 epochs.

Tuning with Optuna

Bayesian Search Parameters

Learning Rate: [1e-5, 5e-5]

Dropout: [0.1, 0.5]

Batch Size: [16, 32, 64]

Optimization Metric: Validation Loss



Winning Config

Learning Rate: 1.15e-5

(Lower than default, ensures stability)

Dropout: 0.11

(Low regularization needed)

Batch Size: 32

(Optimal noise/speed balance)

Training Dynamics

PLACEHOLDER:

Insert W&B Loss Curve Here
(Showing convergence at epoch 2-3)

Observations

Fast Convergence:

Model learned primary features within 2 epochs.

Early Stopping:

Triggered after 3 epochs of no improvement.

Final Val Loss: 0.1996

Test Set Performance

65.1%

Exact Match Accuracy

92.6%

Hamming Accuracy

54.3%

Precision (Macro)

0.36

F1-Score

Key Insight: High Hamming vs. Low F1 indicates class imbalance. The model is biased towards the majority class (Neutral/Joy) and misses minority classes.

Error Analysis

Failure Case 1: Intensity

Input: 'I feel very happy today'

True: Joy

Pred: Sadness (0.24)

Diagnosis: Flat probabilities. The model failed to trigger the 'Joy' neuron strongly enough (>0.5).

Failure Case 2: Structure

Input: 'I got scolded by teacher'

True: Sadness/Fear

Pred: Joy (0.26)

Diagnosis: Context inversion. 'Scolded' is negative, but model output Joy, suggesting insufficient negative samples.

Current Limitations

Class Imbalance

Dataset skewed to 'Neutral'.
Model ignores minority classes.

Context Nuance

Struggles with subtle negations.
Captures keywords but misses direction.

Flat Probabilities

Confidence scores often < 0.3 .
Standard 0.5 threshold is too high.

Future Enhancements

1. Data & Training

Weighted Loss: Use `pos_weight` in BCE loss to penalize missing minority classes.

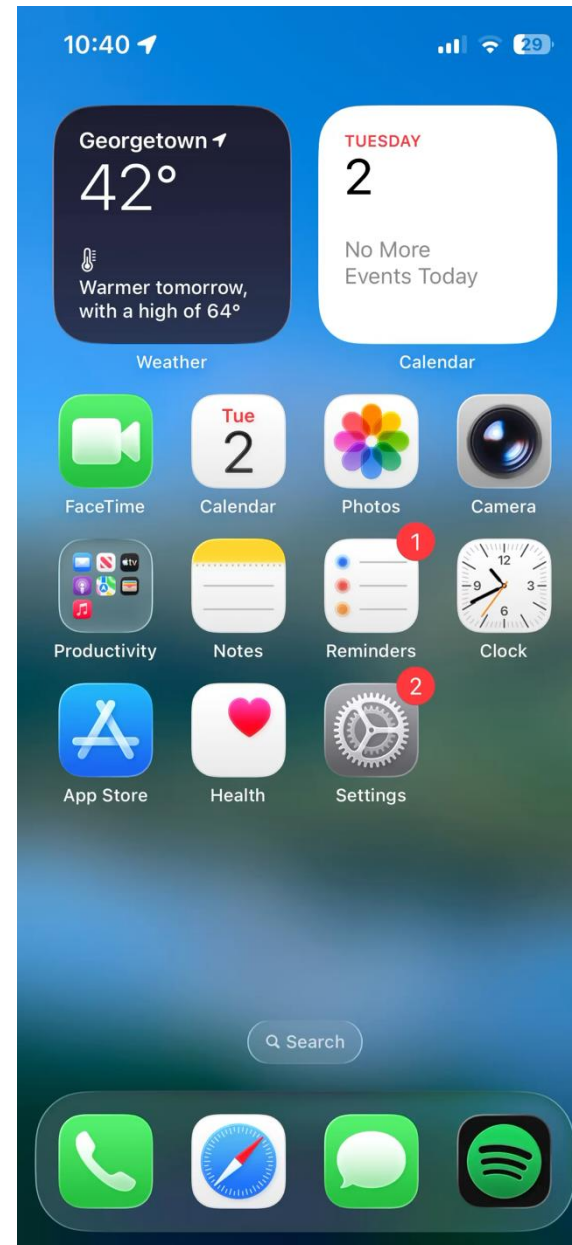
Dynamic Thresholding: Tune thresholds per class (e.g., $\text{Fear} > 0.2$) instead of global 0.5.

2. Architecture

Model Upgrades: Switch to RoBERTa (better pre-training methodology).

Layer Freezing: Train head first, then unfreeze backbone to preserve features.

Demo



Conclusion

Baseline established with 92% Hamming accuracy. Immediate next steps: Class Balancing and Threshold Tuning.